

A certified abstention band for card-fraud decisions, and two quantum arms reported as measured

Global Quantum + AI Challenge 2026 — HSBC track

1 The decision, and what is missing from it

An issuer authorising a card transaction has three actions, not two: approve, decline, or challenge the cardholder through 3-D Secure. The middle one is what makes the problem tractable at scale — routing even one percent of authorisation volume to human analysts is not operationally possible, while a step-up challenge is. Fraud models are reported by AUC or average precision; neither is a statement about a decision, and neither carries a finite-sample guarantee. Conformal methods reach this problem closely — Singh et al. pair class-conditional prediction with cost-controlled abstention, and Zhu et al. certify a false-negative rate on card fraud — but neither conditions the guarantee on the abstention region it routes to, nor evaluates it under a split that orders time. Both are what this submission adds, and the second turns out to matter.

What we contribute. A three-valued rule whose *abstention band* carries a distribution-free, finite-sample risk certificate under a pre-registration frozen before any model was fitted, and an honest measurement of where two quantum approaches sit against a tuned classical baseline on the same decision.

What a successful PoC would demonstrate. A step-up decision whose false-decline rate carries a distribution-free, finite-sample certificate at a level the bank sets, validated on a block it never saw — the artefact a model-risk function can audit, which an AUC cannot provide. This study meets that target on public data, at loose levels only ($\alpha \geq 0.10$) and at the band budgets the sample size permits; the PoC target is the same property at the budget the business chooses rather than the one the data forces.

Expected impact. The deliverable is an operating guarantee, not a score, and three consequences follow from that — none of which needs a quantum result. A model-risk function can review a bound instead of a metric, which is what supervisory revalidation [RG-4] asks of a decisioning model and an AUC cannot supply — and it matters more here, not less, because fraud detection sits in the AI Act’s Annex III *exception* [RG-2], so no external high-risk regime supplies that discipline. The operating point becomes an input: α and the band budget are chosen by the business, and the certificate reports whether that choice is attainable rather than asserting it. And the band bounds what the expensive component costs — Section 5 prices it against the authorisation budget — which is what makes a quantum re-ranker discussable at all. We attach no monetary figure: issuer-side false-decline costs are not public, so the study certifies the *rate* and leaves the bank to multiply it by a number only the bank has.

Neither quantum arm produced a usable result, and they failed differently. A fidelity quantum kernel was rejected by its own a-priori screens before it was run on the task — a decisive negative. A matrix-product-state classifier ran and did not beat gradient boosting, but that comparison is underpowered, so what it yields is a non-superiority bound rather than a null result. We report both in the body rather than the appendix, because a screen that only ever passes is not a screen, and because the pre-registration committed us to reporting whatever it returned.

What we do not claim. No quantum advantage of any kind. No portfolio-level PSD2 compliance figure — the SCA-RTS ceilings govern exemption eligibility, not a decline threshold, and the appendix records that category error as one we made and withdrew. No claim that the certificate binds the *unconditional* false-decline rate: it does not, and Section 2 says why that distinction is the point rather than a caveat.

2 Method

Paradigm. Hybrid classical-quantum: a *gate-based* fidelity kernel and a *quantum-inspired* matrix-product-state classifier, both confined to an abstention band and wrapped in a classical certificate that depends on neither.

The rule. With a classical scorer f and thresholds $\tau_{lo} < \tau_{hi}$ fixed in advance,

$$D(x) = \begin{cases} \text{DECLINE} & f(x) \geq \tau_{hi} \\ \text{STEP-UP, then } g(x) \geq \lambda \Rightarrow \text{DECLINE} & \tau_{lo} < f(x) < \tau_{hi} \\ \text{APPROVE} & f(x) \leq \tau_{lo} \end{cases}$$

The band is $B = \{x : \tau_{lo} < f(x) < \tau_{hi}\}$ and g is the in-band re-scorer, which is where a quantum model would enter if one were usable.

Table 1: The same gradient-boosted model, the same hyperparameters, five seeds, evaluated on the test block of three different splits. A stratified random split inflates average precision by 0.316 over a forward holdout, with non-overlapping ranges. This is the cleanest evidence in the study that a random split does not measure what a deployed fraud model faces. It is one model on one file, so it establishes the size of the effect here rather than a general rule, and it costs nothing to obtain: it is the control arm of the split design, not an extra experiment.

Split	ROC AUC	AP	AP vs forward holdout
stratified	0.9658	0.8254	+0.3164
temporal	0.8851	0.5090	—
card-disjoint	0.8509	0.5394	+0.0304

The estimand is band-conditional, deliberately. We certify

$$R(\lambda) = \mathbb{P}(D(X) = \text{DECLINE} \mid Y = 0, X \in B),$$

not the marginal false-decline rate. The marginal rate is dominated by τ_{hi} and would be almost unchanged if g were replaced by a coin flip. A certificate that does not bind the component it is said to license is not evidence about that component — so the harder, smaller-sample quantity is the one we bound, and the cost of that choice is reported in Section 3 rather than hidden.

Four blocks, and two distinct breaks of exchangeability. $\mathcal{D}_{\text{train}}$ (60%) fits f ; $\mathcal{D}_{\text{band}}$ (10%) is the *only* data that may set the band edges, g ’s hyperparameters or any screen threshold; $\mathcal{D}_{\text{cal}}$ (10%) certifies the composite rule; $\mathcal{D}_{\text{test}}$ (20%) is read once. Blocks are contiguous and snapped to day boundaries.

Two breaks must not be conflated. The **temporal** break is that $\mathcal{D}_{\text{cal}}$ precedes $\mathcal{D}_{\text{test}}$. The **selective** break is that band membership is a data-dependent event, if the edges were estimated on the data used to certify. Freezing the edges on $\mathcal{D}_{\text{band}}$ removes the second by construction: membership becomes a fixed measurable predicate, and filtering a sequence by one preserves exchangeability of what remains. Only the temporal break survives, and we measure it.

Risk control. Learn-then-Test controls family-wise error over a finite grid fixed in advance, using Hoeffding–Bentkus p -values with Holm. Two risks are controlled jointly: $R(\lambda)$ at level α , and the false-negative rate at α_{FN} .

The bound applies only to a *mean of per-observation losses*. Our first formulation was a nonlinear transform of a mean and did not produce valid p -values; it is now a genuine 0/1 loss averaged over frauds, equivalent to a recall floor at $1 - \alpha_{\text{FN}}$.

Coverage is judged against the exact law, not against α . For a sound split-conformal system the test-error count follows $E \sim \text{BetaBinomial}(m, n+1-k, k)$ with $k = \lceil (n+1)(1 - \alpha) \rceil$. Checking $\hat{r} \leq \alpha$ is a one-sided test against the wrong null: on a correctly calibrated system it passes only about half the time. Every verdict below is a two-sided Beta-Binomial interval.

Intervals resample entities, not rows. The IEEE-CIS label propagates a reported chargeback to later transactions sharing an account, email or billing address, so rows are clustered: 85.0% of test-block `card1` values also appear in training. Every interval is a card-level block bootstrap.

3 Results

What a random split buys you, measured. Before any conformal machinery, the split itself. Table 1 runs one gradient-boosted model at fixed hyperparameters over five seeds on three splits of the same file. A stratified random split reports 0.3164 more average precision than a forward holdout — 0.8254 against 0.509 — with ranges that do not overlap across seeds. That is the largest effect in this study, larger than anything either quantum arm could have contributed and visible without any quantum method, and it is why what follows reports a temporal split.

It is also what separates this baseline from the published leaderboard: the 1st-place IEEE-CIS solution reports ROC AUC 0.9459 under time-based cross-validation over the training file, against 0.8837–0.8884 here on a strict forward holdout. Different tasks, and the difference is measured above rather than assumed. The appendix carries the statement’s primary metrics in full, confusion matrix included.

Table 2: Coverage across five seeds per arm and level, judged against the exact Beta-Binomial interval. This replaces a single-split table whose temporal ratio at $\alpha = 0.01$ was the across-seed *maximum* rather than its centre, and which described the card-disjoint arm as inside the interval at every level. The temporal arm breaches on every seed at the two loosest levels; the stratified arm breaches on none at any level.

Arm	α	Ratio (mean)	sd	Seeds outside
temporal	0.010	1.436	0.055	5 of 5
temporal	0.005	1.456	0.036	5 of 5
temporal	0.002	1.314	0.067	4 of 5
temporal	0.001	0.943	0.020	0 of 5
stratified	0.010	1.008	0.038	0 of 5
stratified	0.001	0.962	0.140	0 of 5
card-disjoint	0.010	0.916	0.133	2 of 5
card-disjoint	0.001	0.882	0.362	1 of 5

Grey bands are the exact Beta-Binomial interval; red points fall outside it

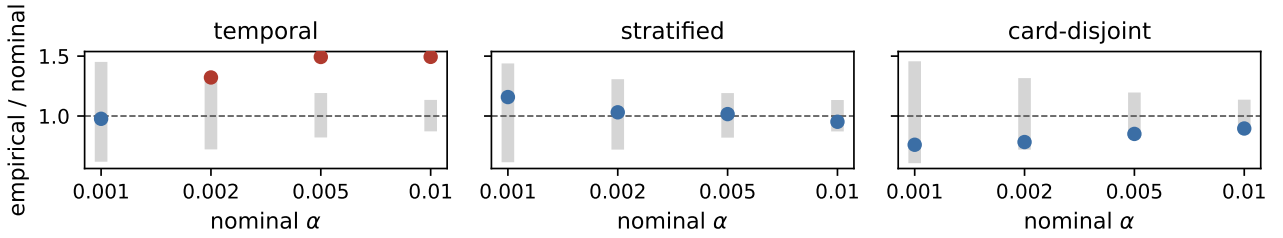


Figure 1: One seed, with the exact Beta-Binomial interval to scale; Table 2 gives all five. The interval widens sharply as α tightens, which is why the temporal arm’s compliance at the tightest level is weak evidence rather than a pass.

Coverage fails under time, on every seed, and the controls do not reproduce it. Table 2 calibrates on $\mathcal{D}_{\text{cal}}$ and reports on $\mathcal{D}_{\text{test}}$, five seeds per arm and level, each judged against the exact Beta-Binomial interval. The temporal arm falls outside on *every* seed at the two loosest levels and on 4 of 5 at the third, and inside on every seed at the tightest; the stratified arm falls outside on none, at any level.

The card-disjoint arm is not clean, and the two levels fail differently. At $\alpha = 0.01$ it is outside on 2 of 5 seeds and both *over-cover* — a two-sided fit failure that leaves the one-sided guarantee intact. At $\alpha = 0.001$ it is outside on 1 of 5, and that seed runs the other way: ratio 1.5245 against four seeds that all sit below one, the lowest at 0.6502. One seed against four reads as noise rather than a systematic breach, but it is under-coverage, so we do not report this arm as a pass. Its seed-to-seed spread (sd 0.362) is an order of magnitude wider than the stratified arm’s. What the controls establish is therefore narrower than an earlier draft claimed, and still sufficient: neither reproduces the temporal arm’s systematic, every-seed breach. That is weaker than “the breach is temporal” — the arms differ in more than time — and it is what three arms can support.

It is not label censoring: the trailing-window fraud rate is 3.666 % against 3.281 % earlier — higher, not lower — with no monotone trend ($\tau = -0.2857$, $p = 0.3988$).

But the breach is origin-dependent, and this narrows the claim. Re-running at five rolling calibration origins gives Table 3: the ratio ranges from 0.684 to 1.411. 2 of 5 breach, and the earliest is *conservative* — the deviation reverses direction. So the defensible statement is not that temporal splits inflate the false-decline rate by a fixed factor, but that the deviation depends on where the calibration window falls, and a single origin cannot establish its size.

Table 3: The temporal deviation across five rolling calibration origins. Two of five breach and the earliest is conservative, so no single ratio characterises the effect.

Origin (day)	Ratio	Errors	Verdict (interval)
101	0.684	379	conservative ([472, 641])
111	0.893	459	inside ([437, 595])
121	1.411	777	breached ([468, 638])
131	1.198	696	breached ([494, 674])
141	1.126	607	inside ([457, 625])

Table 4: H4 in the band: matrix-product-state against the tuned gradient-boosted baseline on identical rows and features. Intervals are 95 % card-clustered bootstraps. Every interval contains zero, and the comparison is underpowered against its pre-registered ceiling, so what is established is a non-superiority bound rather than a null.

χ	AP (MPS)	AP (GBDT)	Δ AP	95 % CI	p
4	0.1202	0.1407	-0.0205	$[-0.0593, +0.0149]$	0.864
8	0.1173	0.1407	-0.0234	$[-0.0630, +0.0122]$	0.895
16	0.1183	0.1407	-0.0225	$[-0.0613, +0.0136]$	0.884
32	0.1235	0.1407	-0.0172	$[-0.0582, +0.0200]$	0.806

And it holds on the held-out block. This is H5. All 5 certified configurations were applied unchanged to $\mathcal{D}_{\text{test}}$ — same band edges, same λ , no recalibration — and 5 of 5 realised a band-conditional false-decline rate at or below the α they were certified at, the tightest using 0.856 of its budget. Holding a ceiling that loose is a real check that the machinery transfers and a weak one, which is what amendment A1 predicted from the band’s sample size before the arm ran.

What the certificate actually reaches. 5 of 48 configurations certify, all strictly inside the band — the smallest margin below τ_{hi} is 0.0136. That is not automatic: a certificate whose λ sits at the band edge flags nothing, so its risk is structurally zero and the guarantee is vacuous. An earlier version of this table was exactly that, and the margin above is what rules it out.

The survivors need $\alpha \geq 0.10$ and $\alpha_{\text{FN}} = 0.45$, and nothing certifies at the tightest band budget — where $\mathcal{D}_{\text{cal}}$ contributes only 847 legitimate in-band rows against 58343 overall. The binding mechanism is sample size acting through the concentration bound, not the class-conditional degeneracy floor.

4 Two quantum arms, and how each failed

The kernel never reached the task. Before running a fidelity quantum kernel we pre-registered screens over 120 configurations of encoding $\times$ qubits $\times$ bandwidth $\times$ entanglement, and gated on two: a kernel is usable only if it is not exponentially concentrated — effective rank $r_{\text{eff}} = (\sum_i \sigma_i)^2 / (n \sum_i \sigma_i^2)$ inside a usable band — *and* not reproducible by a tuned RBF, $\rho_{\text{RBF}} < 0.60$.

28 of 120 pass conditioning. **0 pass both.** The closest any configuration came to distinctness was $\rho_{\text{RBF}} = 0.6291$; the median correlation is 0.974 and the maximum 1.0, indistinguishable to floating point. On 300 stratified rows with at most eight features, every kernel we could construct is to within measurement a classical RBF in disguise — which we would not have known had we reported a downstream accuracy instead. One caveat: the protocol names the Huang geometric difference first and the implementation computes it without gating on it (amendment A6); since Huang treats a *large* difference as favourable, gating would have made this rejection less decisive, not more. The closest published comparator agrees: a controlled benchmark on real card-transaction data (Faryad, arXiv:2608.15718, 2026) reports no robust quantum advantage, and finds one apparent advantage to be an artefact of unequal search budgets.

The tensor network ran, and did not win. A matrix-product-state classifier contracts $f(\mathbf{x}) = \sum_{\{s\}} A^{s_1} \dots A^{s_N} \prod_j \phi^{s_j}(x_j)$ with $\phi(x) = [\cos \frac{\pi x}{2}, \sin \frac{\pi x}{2}]$, sweeping the bond dimension χ rather than tuning it. In the band, on identical rows and features, it reaches average precision 0.1235 at the largest bond dimension against 0.1407 for the tuned gradient-boosted baseline, a paired difference of -0.0172 .

At full scale it loses by a wider margin, with replication behind it. On all 431 features, trained on $\mathcal{D}_{\text{train}}$ and scored once on $\mathcal{D}_{\text{test}}$, across 16 fits — four bond dimensions at four seeds each — the best draw reaches 0.2512 average precision against 0.5055–0.5114 for the tuned baseline. Every one of the 16 loses by a factor of two or more at a 3.412 % positive rate, so which draw is reported does not change the conclusion.

Capacity is not resolvable, and replication is how we know. The largest spread across seeds at a fixed bond dimension is 0.1781 average precision; the spread of the seed *means* across bond dimension is 0.0629 — seed noise is 2.83 times the capacity signal, so a per- χ ordering is selection on a maximum. **And 2 of 16 fits never left chance**, both at one seed and at two different bond dimensions, on the sequential contraction — the stable path. Failure to train at that rate is a property of this ansatz on this data, and a reader choosing a method for a payment system should be told.

The sweep is affordable because cost does not follow the bond dimension the way the ansatz suggests: the jobs took 2747 to 3145 seconds against the 64-fold spread a χ^2 model predicts. A 431-site chain is bound by the launch overhead of its sequential contractions, not their arithmetic, so capacity is nearly free and depth is the cost.

The comparison is underpowered, and the gate that should have said so was itself wrong. It is a comparison between model *families*, whose measured standard error is 0.0199 — 8.03-fold the baseline-against-itself figure the gate originally used. The effect it can resolve is 0.0557 average precision, against a pre-registered ceiling of 0.023; the appendix gives both defects in full. So H4 is reported as underpowered, which is what the protocol commits to. Every interval contains zero, so any improvement is bounded above by +0.02 average precision — a non-superiority bound, not evidence of equivalence and not evidence that the tensor network is worse.

5 Feasibility, resources, and the constraints we already know

What the PoC sprint would build. The same protocol on the issuer’s own authorisation stream: fit the incumbent scorer, freeze four blocks, calibrate the band, certify the band-conditional false-decline rate at a level the business picks. Every component exists and has been run end to end here, so the sprint is a port and a re-calibration rather than a build: three stages of roughly a month with a month of slack — split integrity, then calibration and certification, then the quantum re-ranker, which is deliberately last because it is the only stage that can fail to produce a result.

Compute, measured rather than estimated. The classical core is cheap: the tuned baseline fits the in-band problem in 0.28 seconds, and conformal calibration is a sort over the calibration block. The expensive component is the quantum-inspired arm, and its cost is known — 16 full-scale tensor-network fits over 356216 training rows and 431 sites took 2747 to 3145 seconds each on the one RTX 6000 PRO Blackwell (96 GB), CUDA 13, Python 3.12 workstation every number here was measured on. Because per-step cost is flat in the bond dimension (Section 7), capacity is nearly free and chain length is the cost, so a resource estimate scales with the feature count and not with the ansatz. No QPU is required: both arms are simulator-only, which the statement permits, and the screens ran on 300 stratified rows across 120 configurations.

Latency, against the budget the statement gives. Of the 300 ms authorisation ceiling about 130 ms is network, leaving roughly 170 ms issuer-side. Scoring one authorisation costs 0.0841 ms, 0.3197 ms at the tail; one in-band kernel evaluation would cost 96.577 ms — 1148 times as much, its tail taking 75.9 % of that budget. The classical core is therefore not the latency risk and the kernel is: it fits only because the band rations it. Both figures bound model cost on this workstation, single-threaded, rather than production latency.

Data. The method needs settled labels — a chargeback window closed before the block boundary — and an entity key to cluster on. Both are ordinary in an issuer’s data and only approximated here.

Constraints we have measured, and would carry into Phase II. The population is conditional on an incumbent: this file contains only transactions an existing stack approved, because declined attempts cannot generate a chargeback. That is exactly the population a champion/challenger comparison runs on, so it is the right restriction — but no result extrapolates past it. The label is partly an entity flag, since a chargeback propagates to later transactions sharing an account, email or billing address, and Shapley attribution makes that concrete: 0.758 of the in-band model’s total contribution falls on `card1` alone, across 8 features. A governance review would read that as a model which mostly recognises cards — the strongest argument for running this on an issuer’s own data, where an entity key is a key and not a predictor. The certificate is sample-starved by design: conditioning on a narrow band is what makes the guarantee bind the component we care about, and also what leaves 847 legitimate rows to certify with. A larger calibration block or a wider band buys tighter α ; the method is unchanged.

What would change the quantum conclusion. An encoding demonstrably outside the RBF family, at a bandwidth that does not concentrate, would pass the same screens unmodified — we did not find one. For the tensor network the honest requirement is a comparison powered for cross-family noise: resolving an effect the size of the interval in Section 4 needs about 7.77 times the in-band evaluation set, which a wider band or a longer window would supply. And the negative is about aggregate discrimination, not the operating point — the screens read the Gram spectrum and H4 compares average precision across the band, while H2, which would have tested decisions at the certified threshold, never ran. A method that does not improve ranking can still improve a decision at a low false-positive threshold, which is the regime this certificate operates in and which this study did not measure.

6 Validation plan

The plan below is not aspirational. Every step of the certification pipeline — split, band, calibration, certificate, held-out validation — was executed in this study on public data, and the results are in Section 3; what a PoC sprint changes is the data, not the protocol. Some pre-registered items were *not* executed, and are named here rather than left to be discovered: the two kernel hypotheses and the entanglement ablation, all closed by the screens before they could run, and the drift measurements recorded in amendment A7, for which no script was written. The appendix enumerates both. A validation plan that cannot say which of its own steps ran is not a plan, and it is why the negative quantum results are in the body rather than buried.

The success metric is a certificate, not a score. A PoC succeeds if the band-conditional false-decline rate

is certified at a level the business names, with family-wise error control, and the certificate then holds on a block that was never used to build it. Average precision and ROC AUC are reported alongside, but they decide nothing: they are not statements about a decision and they carry no finite-sample guarantee. The corresponding failure is equally well defined — the admissible set comes back empty, which is a real outcome and is what happens here at the tightest band budget.

Held-out validation is the acceptance test. The certificate is selected on calibration data and then read once on a fold protected by a persisted ledger that refuses a second distinct configuration. In this study 5 of 5 certified configurations held their risk on that fold, the worst realised risk being 0.856 of its budget. A PoC would run exactly this test and report it whichever way it came out.

Benchmarking is against a tuned baseline, replicated, with the power stated first. The comparator is a tuned gradient-boosted model, not a straw man, fitted on identical rows and features. Intervals resample entities rather than rows, because transactions on one card are not independent. Crucially, a detectable-effect ceiling is fixed *before* the comparison: here 0.023, against an effect the comparison could actually resolve of 0.0557, computed afterwards from its own bootstrap standard error. That gap is why H4 is reported as underpowered rather than as a null. A PoC would size its evaluation block to that ceiling first, which is the single change that would make a quantum-versus-classical comparison on this problem informative.

What keeps the plan honest. Hypotheses, grids, splits and decision thresholds are frozen and hashed before any model is fitted, and a gate refuses undocumented change; multiple comparisons carry Holm correction; coverage is judged against the exact law of Section 2 rather than against α . Every retraction is recorded — this submission carries 68 decision entries, including the ones that removed results we had already written down — and reproduction is `make smoke && make reproduce && make check`.

7 Where the quantum method sits, and why there

Classical, on all traffic. A gradient-boosted scorer runs on every transaction — the only component that sees full volume, and deliberately the strongest available baseline rather than a straw man, at 0.8837–0.8884 ROC AUC over five seeds. A hybrid design whose classical half is weak proves nothing about its quantum half.

Conformal, on the decision. The certificate wraps whatever scores the band, so the deliverable an issuer can act on survives a negative quantum result — as it had to here.

Quantum, on the band only. The abstention band is a few percent of traffic, which is why a quantum re-ranker is conceivable here and nowhere else in the pipeline. At a 3.412% positive rate and a 3-D Secure challenge budget the band holds thousands of rows rather than hundreds of thousands, and a kernel whose Gram matrix costs $O(n^2)$ circuit evaluations is affordable on thousands. The band is not a convenience; it is the only place the arithmetic works.

The compute path. Both quantum arms run on simulators, which the statement permits and does not penalise: no number in this proposal comes from a QPU. The fidelity kernel’s reference implementation is an exact statevector, with Braket’s `LocalSimulator` and Aer on CPU and GPU agreeing to within 3.442e-15 over 12 comparisons. The tensor network runs in PyTorch on a single GPU, bound by kernel-launch overhead rather than arithmetic (Section 4). On the sample count the statement asks for: the screens ran on 300 stratified rows per configuration across 120 configurations, and the kernel arm was rejected before any downstream evaluation, so no larger quantum execution took place. The circuits are within near-term reach — at most 8 qubits, transpiled depth 35 and 28 two-qubit gates on a portable basis — so what stopped this arm was the screens, not the hardware.

8 Team

Amon Koike — Quantum Research Engineer, registered sole proprietor in Japan, remote-first. B.S. Physics, Tokyo Denki University. IBM Qiskit Advocate (2026). Single-person team. Portfolio and code: `thedaemon-wizard.github.io/portfolio-quantum-research-engineer`.

The statistical machinery in this proposal is not new to the author. In QIntern 2026 Project 12 (QWorld), within Team A — which owns the paper’s conformal-coverage proposition — he wrote the split-conformal calibration of the false-alarm rate and the results freeze, behind a SHA-256 manifest with the 135 checks green at that freeze. The team’s harness covers three intrusion-detection corpora at five seeds, with exact Beta-Binomial coverage and McNemar under Holm. *Protocol-final and numbers-provisional in the team’s own words — the quantum side runs on a placeholder score interface — so what it establishes is machinery, which is what this submission reuses.*

Directly relevant prior results. *Tensor networks:* Yale Peaked Hackathon 2026, 13th of 549 at 450 of 550, by his own matrix-product-state marginal attack on circuits to 69 qubits. *Quantum kernels:* QPoland 2025 runner-up. *Upstream:* unitaryHACK 2026, merged into CUDA-Q and QuEST. *Hardware:* a published QML benchmark run on IBM `ibm_fez`.